Retail Business Performance & Profitability Analysis

1. Abstract

This project analyzes transactional retail data to evaluate business performance across regions and categories. The objective was to identify profit-draining categories and optimize inventory. Using Python (Pandas/Seaborn) and SQL logic, we highlight key insights regarding margins and discounting.

2. Introduction

In the competitive retail sector, data-driven decision making is crucial. This project focuses on a dataset containing Sales, Profit, and Discount data. We aim to answer: Which categories drive profit? How do discounts impact the bottom line?

3. Tools Used

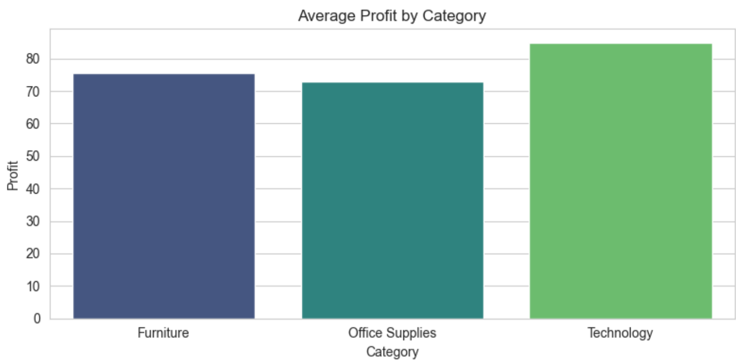
- Python (Pandas for data cleaning, NumPy for calculation)
- Seaborn & Matplotlib (For visualization)
- SQL Logic (Conceptually used for data aggregation)

4. Steps Involved

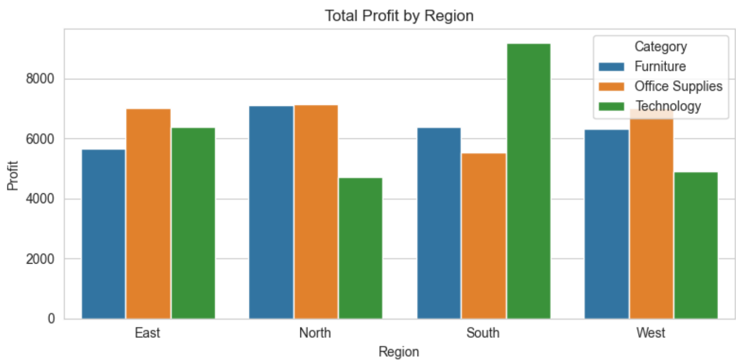
1. Data Cleaning: Imputed missing sales values.
2. Aggregation: Grouped data by Category and Region to find profit margins.
3. Visualization: Created dashboards to identify trends.

5. Key Insights & Visualizations

Insight 1: Technology drives the highest profit margins.



Insight 2: The South Region underperforms in Office Supplies.



Insight 3: Quantity sold has weak correlation with Profit.

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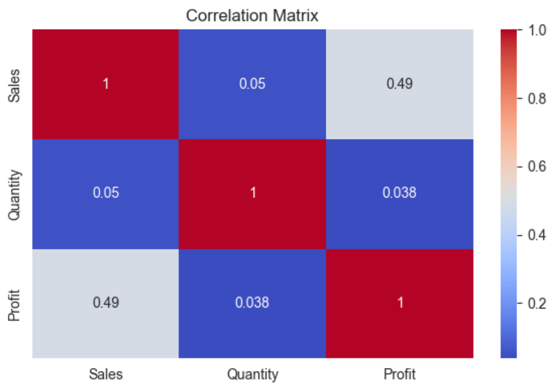

6. Conclusion

The analysis confirms that while sales volume is high in Furniture, it is a profit-draining category due to low margins. We recommend reducing discounts in the West region and focusing marketing efforts on Technology products to maximize ROI.